Rerunning Flaky Tests under Controlled Variations

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Flaky tests pass and fail on the same code. This paper reruns each failing test under controlled variations.

CCS Concepts: • **Software and its engineering → Software testing and debugging**.

Additional Key Words and Phrases: flaky tests, regression testing

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1 Introduction

Continuous integration runs the test suite of a project after every change to the code. A failing test stops the merge of the change until a developer finds the cause of the failure and repairs the code or the test. Most failures point to a fault in the change itself. Some failures, however, come from tests that pass and fail on the same code, and these flaky tests cost developers hours of work on faults that do not exist.

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The detector of this paper reruns each failing test under controlled variations of its environment. It changes the order of the tests, the timing of the threads, and the seed of every random number generator. A test whose outcome changes under any variation is flaky, and the variation that changes the outcome names the likely cause. The detector reports the test, the variation, and the lines of the test that read the shared state.

We evaluate the detector on the failures of twelve Java projects over one year of builds. The projects range from small libraries with a few hundred tests to large frameworks with tens of thousands of tests. For each failure, the build logs record the commit, the failing tests, and the outcome of the next build on the same commit. A failure that the next build does not repeat is a candidate flaky failure, which two authors then label by hand.

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Developers accepted most of the repairs that the detector suggested. A repair that adds a condition to a wait or a reset of a shared field takes a few lines of code and leaves the intent of the test intact. Repairs of tests that depend on the network were less popular, since they replace a real service

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Several threats limit these results. The labels of the failures come from two authors, who agreed on most cases and resolved the rest in a discussion. The projects use Java and Maven, and other languages and build tools may show other causes of flakiness. Finally, one year of builds may miss causes that appear only after a change of the build machines, such as a new version of the operating system or of the virtual machine. Continuous integration runs the test suite of a project after every change to the code. A failing test stops the merge of the change until a developer finds the cause of the failure and repairs the code or the test. Most failures point to a fault in the change itself. Some failures, however, come from tests that pass and fail on the same code, and these flaky tests cost developers hours of work on faults that do not exist.

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References

[1] A. Author. 2020. A study of flaky tests. In *Proc. Conf.* 1–10.